■# IBI Bank — PBL Presentation Outline

Suggested presentation: 10–12 slides

Slide 1 — Title

• Banking and Digital Transaction Management System

• IBI Bank — International Bank of India

• Course / subject code: BTY39105

• Team name and member names

• Faculty guide and institution

Slide 2 — Problem Statement

• Banking operations involve related customer, account, transaction, beneficiary, and loan information.

• Reliable storage and retrieval are essential for consistency and reporting.

• The project demonstrates a database-backed banking management application.

Slide 3 — Objectives

• Maintain related banking records in MySQL.

• Support authenticated customer and administrator workflows.

• Record transfers and transactions.

• Manage loans and EMI payment information.

• Demonstrate SQL queries and advanced database objects.

Slide 4 — Technology Stack

• Front end: HTML, CSS, JavaScript

• Backend: Node.js and Express

• Database: MySQL

• Authentication and verification: describe the mechanisms actually implemented

• Development tools: list the tools actually used by the team

Slide 5 — System Architecture

Show the flow:

Customer/Admin → Web Interface → Express API → MySQL Database → Results returned to the interface.

Mention that backend authorization and validation should protect sensitive operations.

Slide 6 — ER Diagram

Insert the final ER diagram from Documentation/er_diagram.mmd after rendering it.

Explain major entities and relationships: Customer, Account, Branch, Account Type, Transaction, Fund Transfer, Beneficiary, Loan, and EMI Payment.

Slide 7 — Database Design and Normalization

• Explain primary keys and declared foreign keys.

• Summarize 1NF, 2NF, and 3NF.

• Explain why customer, account, transaction, loan, and EMI data are stored separately.

• Use the final schema to support claims about constraints and relationships.

Slide 8 — SQL Implementation

Demonstrate examples from:

• Basic SELECT queries

• Aggregate functions and GROUP BY

• JOIN queries

• Subqueries and correlated subqueries

• Views, indexes, stored procedures, function, and triggers

• Transaction control using COMMIT and ROLLBACK

Slide 9 — Application Features

Use screenshots of implemented pages:

• Login and email verification

• Customer/account management

• Beneficiary and transfer management

• Loan and EMI management

• Transaction history and reports

Only include features that are present and working in the submitted version.

Slide 10 — Testing and Results

• Show the testing report and actual outcomes.

• Include screenshots or output evidence.

• Explain any issues found and how they were resolved.

• Do not report unperformed tests as successful.

Slide 11 — Limitations and Future Enhancements

Choose realistic enhancements, such as:

• Improved audit and monitoring reports

• More automated integration tests

• Additional reporting filters

• Deployment and backup improvements

Include only limitations that apply to the current implementation.

Slide 12 — Conclusion and Contributions

• Summarize the database and application objectives achieved.

• State each team member's actual contribution.
• Include repository and live-demo URLs only if they exist and are accessible.
• Finish with questions.

Presentation preparation checklist

• Replace every placeholder with accurate project information.
• Add screenshots from the real application.
• Insert the final ER diagram.
• Include representative SQL examples and outputs.
• Verify that the presentation and report agree with the final database schema.